

CHALLENGE BRIEF

The modern Energy Consultant

How might we give anyone a clear, personal plan for making their home more energyefficient. Without needing an expert or a big budget to get started?

2. PARTNER

Kenergy Solutions GmbH AI-powered energy startup from the Rhein-Main region. Kenergy helps homeowners and solar installers unlock the potential of solar energy: their platform analyzes every rooftop in a supply area via satellite data and automatically generates an individual solar offer for each property. No manual effort required. www.kenergy-solutions.de

3. THE PROBLEM

Most people genuinely want to live more sustainably at home. They know they could save money on heating and electricity, reduce their environmental footprint. But when they actually try to figure out where to start, it gets way to time and money intensive for them. The information out there is either too generic or too complicated. A quick online search returns broad tips that feel irrelevant to your specific situation: a rented flat on the third floor of a 1970s building is very different from an owner-occupied house built in 2005. Going the professional and traditional way means hiring a certified energy consultant, which is expensive, slow and sometimes maybe to intimidating for people. The result: people who want to act, don't. The knowledge gap is the bottleneck, not the willingness.

4. THE OPPORTUNITY

Imagine a digital platform that works like a smart friend who happens to know everything about energy and sustainability. You tell it about your home: where you live, what kind of building it is, how you heat it, whether you own or rent. And it tells you exactly what you can do, what it will cost, what you'll save, and how to get started. Not a list of generic tips. A real plan, built for your situation.

Kenergy Solutions already does something close to this for solar: they use AI and satellite data to generate individual rooftop analysis and offers, automatically, at scale. The next frontier is expanding this logic to the full picture of residential energy: heating, insulation, cooling, appliances, behavioral changes. Making it accessible to everyone, whether they own their home or rent it.

The market for home energy renovation in Europe is enormous, and government subsidies are sitting unclaimed because people simply don't know they qualify. A platform that bridges this gap doesn't just create a business opportunity. It directly accelerates the energy transition, one household at a time.

Compared to Kenergy Solutions, we do not expect that an output delivers a complete and detailed offer and planning of each energy system. We rather expect a solution that provides guidance and rough (but accurate) estimates and then can refer you to a specific installer (e.g., solar).

5. WHAT THE PARTNER PROVIDES

Kenergy Solutions provides the following to teams working on this challenge:

Resource	Details
Building data	TABULA/EPISCOPE: a European database of residential building types organized by construction period, including typical insulation levels and heating demand (episcope.eu). ETHOS.BUILDA: an open dataset covering all residential buildings in Germany with construction year, size, and energy archetype (zenodo.org/records/12069755).
Domain mentor	An energy and product expert from Kenergy Solutions, available for a 20-minute kickoff briefing and two 1-hour Q&A windows during the hackathon.

6. CONSTRAINTS

- The outputs should be individualized and personalized to each user.
- In contrast to most other (digital) energy consulting platforms, the user should get a direct response to his questions/situation. On different platforms, you always have to fill out forms and wait for them to get back to you.
- The platform must serve both renters and homeowners. These two groups have very different options. A renter cannot install a new heating system, but they can switch tariffs, change appliances, or have a conversation with their landlord. A solution that only works for property owners misses the majority of the population.
- The solution should work for people with no technical background. The target user is not an engineer. The platform should handle complexity, not the user.

7. DELIVERABLES

Every team presents the standard Build Days deliverables:

- Problem-customer pair: Who exactly are you building this for, and what did you learn about their situation during the hackathon? Show us real insight, not assumptions.
- Solution concept with clear value proposition: What does your platform do, for whom, and why is it better than what people do today? Make the core idea easy to grasp.
- Tangible demonstration: Show us something real. A working prototype, a clickable mockup, a simulation, or a live demo. The format is up to you; the experience is what matters.

The focus lies on the solution and its tangible demonstration.

8. EVALUATION CRITERIA

Standard Build Days criteria apply across all challenges:

Criterion	Description
Problem Validation	Did the team identify a specific customer, validate a real pain point, and show evidence?
Solution Quality	Is the concept original, well-reasoned, and clearly articulated?
Prototype Tangibility	Did the team move beyond slides to something concrete and demonstrable?
Business Potential	Could this become a real venture? Is there a market, a model, a path forward?
Presentation & Team	Is the presentation clear, compelling, and delivered with confidence?

Partner bonus criterion: Personalization quality. Does the platform give meaningfully different and genuinely useful guidance depending on who is asking?

9. RESOURCES & INSPIRATION

Starting points to help you get oriented. Not a reading list!

Kenergy Solutions The challenge partner's own platform shows what AI-driven, individualized energy analysis looks like in practice. A good reference for the product direction. www.kenergy-solutions.de

TABULA European Building Typology Standard reference for residential building types and their energy characteristics across Europe. episcope.eu

ETHOS.BUILDA Open dataset of all German residential buildings: construction year, type, and energy archetype. zenodo.org/records/12069755

Bright Sky API Simple, free API for German weather data (temperature, solar irradiation, heating degree days by location). brightsky.dev

co2online Heizspiegel National heating benchmark tool. Useful both as a data source and as an example of what a consumer-facing energy tool looks like today. co2online.de

BAFA & KfW Official sources for Germany's home energy subsidy programs. bafa.de and kfw.de

Enter & Deutsche Sanierungsberatung Two early-stage startups working adjacent to this problem space. Worth understanding where they focus and what's still missing. enter.de